MILITARY SPECIFICATION

SIMULATOR ASSEMBLY

WARHEAD SECTION: M3E3

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of simulator entitled Simulator Assembly, Warhead Section: M3E3 which is used on the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8857917 - M3E3 Simulator Warhead Section.
EL-8854178 - Equipment List.

FSC: 1115

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The simulator assembly shall comply with all requirements specified on the drawing (dwg) 8857917, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Operational characteristics.

3.5.1 Insulation resistance. The insulation resistance between the pins of connector P919, as designated in Table I shall be 20 megohms minimum when tested as specified in 4.4.1.

Table I

From Pin	To Pins
4	9,13,15,24,35,36,50,62,Case
9	13,15,24,35,36,50,62,Case
13	15,24,35,36,50,62,Case
15	24,35,36,50,62,Case
24	35,36,50,62,Case
35	36,50,62,Case
36	50,62,Case
50	62,Case
62	Case

3.5.2 Circuit resistance and continuity. The resistance between connector pins shall be as designated when the stepping switch is in the position indicated in Table II. Pulse the stepping switch with 28 volts direct current (vdc) plus or minus (+) 10 percent with a pulse width of less than one second. The positive pulse shall be applied to P919-31 and the return shall be connected to P919-18. Test as specified in 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Table II

Switch Position	Pin	To	Pin	Resistance (Ohms)	Tolerance (+ percent)
	P919-36		J14-E	0.2 max	
	P919-9		J14-B	0.2 max	
	P919-9		P919-42	0.2 max	
	P919-9		P919-3	0.2 max	
	P919-21		P919-14	0.2 max	
	P919-21		P919-27	0.2 max	
	P919-21		P919-40	0.2 max	
	P919-21		P919-1	0.2 max	
	P919-21		P919-8	0.2 max	
	P919-24		P919-31	0.2 max	
	P919-50		P919-70	0.2 max	
	P919-28		P919-35	30,000	10
	P919-15		P919-22	715 **	10
	P919-18		P919-24	10 min **	
	P919-11		P919-4	290 **	10
	P919-11		P919-17	440 **	10
	P919-62		P919-68	11,000	0.1
1 *	P919-50		P919-69	3,000	0.1
2 *	P919-50		P919-68	3,000	0.1
3 *	P919-50		P919-68	5,000	0.1
4 *	P919-50		P919-68	7,000	0.1
5 *	P919-50		P919-68	9,000	0.1
6 *	P919-50		P919-68	9,000	0.1

* Apply 28 vdc pins to pins P919-31 and P919-18 to pulse ledex.
** Measure at 80° ± 20°F. only.

3.5.3 Operation test. Each simulator assembly shall be operated as specified in 4.4.3.

3.6 Environmental testing. The simulator assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The simulator assembly shall also comply with 3.5 before and after each environmental test. Testing shall be as specified in 4.4.1, 4.4.2, 4.4.3 and 4.4.4. Tests shall be conducted in the order specified.

3.6.1 High temperature. The simulator assembly shall show no evidence of damage or deterioration upon exposure to 160° ± 5 degrees Fahrenheit (°F) and while at this temperature shall comply with 3.5 when tested as specified in 4.4.4.1.

3.6.2 Low temperature. The simulator assembly shall show no evidence of damage or deterioration upon exposure to -65 ± 5°F and while at this temperature shall comply with 3.5 when tested as specified in 4.4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

3.6.3 Transportation vibration. The simulator assembly shall be subjected to vibration as shown in Table III. The total duration of the test in each of three mutually perpendicular axes shown in Figure I shall be 15 minutes minimum at +80 ± 20°F. No evidence of damage affecting function shall result when tested as specified in 4.4.4.3. Following this test, the assembly shall meet the requirements of 3.5.1, 3.5.2 and 3.5.3.

Table III

Frequency Range (Hz)	Time Cycle	Constant Acceleration or Double Amplitude
8-55-8	7.5 min.	0.06 in ± 0.006 in.
55-500-55	7.5 min.	10 g ± 1 g

3.6.4 Shock. The simulator assembly shall withstand shocks of 30 ± 5 gravity (g) units three times in the direction along the three mutually perpendicular axes shown in Figure I for a total of 18 shocks when tested as specified in 4.4.4.4. The duration of shock shall be 11 ± 2 milliseconds (ms) as measured between points which are 10 percent of the peaks. The time to peak shall be 5.5 ± 1 ms. Following the tests, the assembly shall meet the requirements of 3.5.1, 3.5.2 and 3.5.3.

3.7 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 8857917) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Insulation resistance (see 3.5.1)		X
3. Circuit resistance and continuity (see 3.5.2)		X
4. Operation test (see 3.5.3)		X
5. High temperature (see 3.6.1)		X
6. Low temperature (see 3.6.2)		X
7. Transportation vibration (see 3.6.3)	X	
8. Shock (see 3.6.4)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Simulator assembly, warhead section: M3E3 (see dwg. 8857917) before potting.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Loose connection or incorrect soldering	Visual/Manual
102.	Damaged or missing components	Visual
103.	Damaged wire insulation	Visual
104.	Defective casting	Visual
105.	Incorrect orientation of connector receptacles	Visual
106.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Loose mounting bolts or screws	Gage*/Visual
203.	Improper protective finish and bare spots	Visual/Manual

*Do not gage screws that have visual indication of having been sealed with seal and retaining compound (Loc-tite).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

4.3.2.1.2 Simulator assembly, warhead section: M3E3 (see dwg. 8857917) after potting and prior to securing cover assembly.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Damaged component	Visual
	102. Damaged wire insulation	Visual
	103. Mounting bolts or screws missing	Visual
	104. Presence of potting on potentiometer bearing	Visual
	105. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Insufficient potting	Visual
	202. Improper protective finish	Visual
	203. Evidence of poor workmanship (see 3.8)	Visual

4.3.2.1.3 Simulator assembly, warhead section: M3E3 (see dwg. 8857917) after securing cover assembly.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location of mounting holes	Gage
	102. Defective cover assembly	Visual
	103. Missing cover screws	Visual
	104. Damaged or missing nameplate	Visual
	105. Bent pins in connector	Visual
	106. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Loose cover screw	Gage*/Visual
	202. Improper information on nameplate	Visual
	203. Evidence of poor workmanship (see 3.8)	Visual
	204. Serialization improper or missing (see 3.4)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

*Do not gage screws that have visual indication of having been sealed with seal and retaining compound (Loc-tite).

4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation packaging and packing not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified below:

Test	Test Classification
a. Insulation resistance	Major
b. Circuit resistance	Major
c. Operation test	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Insulation resistance, circuit resistance and operation test. Each simulator assembly shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.5.1, 3.5.2 and 3.5.3, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854178). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. Insulation resistance shall be measured by means of an insulation resistance tester at 500 ± 50 vdc. The maximum current output of the insulation resistance tester shall not exceed 5 Milliamperes. (All switches must be on position 1 unless otherwise specified). The insulation resistance shall be measured between those pins specified in 3.5.1. Requirements shall be as specified in 3.5.1.

4.4.2 Circuit resistance. A suitable electrical resistance measuring instrument shall be connected to the connector pins specified in 3.5.2 for the measurement of circuit resistance.

4.4.3 Operation test. Step the stepping switch through at least one complete revolution. The switch shall be stepped and monitored as indicated in Table II.

Connect a 680 ohm ± 5 percent, 5 watt resistor to pins P919-8 and P919-11. Connect a Bendix Barosetter type 8821748 (see Figure 2), to the appropriate terminals on P919. Set the Baro Setter at -1300 feet. Press the setting button until a null is indicated (a null is indicated on the Bendix Barosetter when the green "null" light comes on). Repeat the above for settings of 10,000 feet and 25,000 feet. Measure the voltage between pins P919-8 (+) and P919-11 (-). This voltage shall measure 46 ± 6 vdc as long as the barosetter is connected and power is applied.

CAUTION: BE SURE 680 OHM RESISTOR IS CONNECTED TO PINS P919-8 AND P919-11 BEFORE ANY POWER IS APPLIED.

Disconnect Bendix Barosetter. Connect the positive side, 200 vdc ± 10 percent to pin P919-8. Connect a 100K ohm 1/2w ± 5 percent resistor from pin J14-A to the negative side of the 200 vdc. Measure the voltage across the 100K resistor. The measured voltage across the 100K resistor shall not exceed 1.0 volt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

4.4.4 Environmental testing (see 3.6).

4.4.4.1 High temperature. The simulator assembly shall be placed in a temperature chamber and the temperature raised to 160 ± 5°F. and held at this temperautre for a minimum of four hours after stabilization (see 6.4). While at 160 ± 5°F. test according to 4.4.1, 4.4.2 and 4.4.3. Requirements shall be as specified in 3.6.1.

4.4.4.2 Low temperature. The simulator assembly shall be placed in a temperature chamber and the temperature lowered to -65 ± 5°F. and held at this temperature for a minimum of four hours after stabilization (see 6.4). While at -65 ± 5°F. test according to 4.4.1, 4.4.2 and 4.4.3. Requirements shall be as specified in 3.6.2.

4.4.4.3 Transportation vibration. The simulator assembly shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes, with the fixture mounted on a suitable vibration producing apparatus. After vibration, test according to 4.4.1, 4.4.2 and 4.4.3. Requirements shall be as specified in 3.6.3.

4.4.4.4 Shock. The simulator assembly shall be rigidly mounted in a fixture which is capable of being securely mounted in a suitable shock producing device, and shall be shocked as specified. After shock, test according to 4.4.1, 4.4.2 and 4.4.3. Requirements shall be as specified in 3.6.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The simulator assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The simulator assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The simulator assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

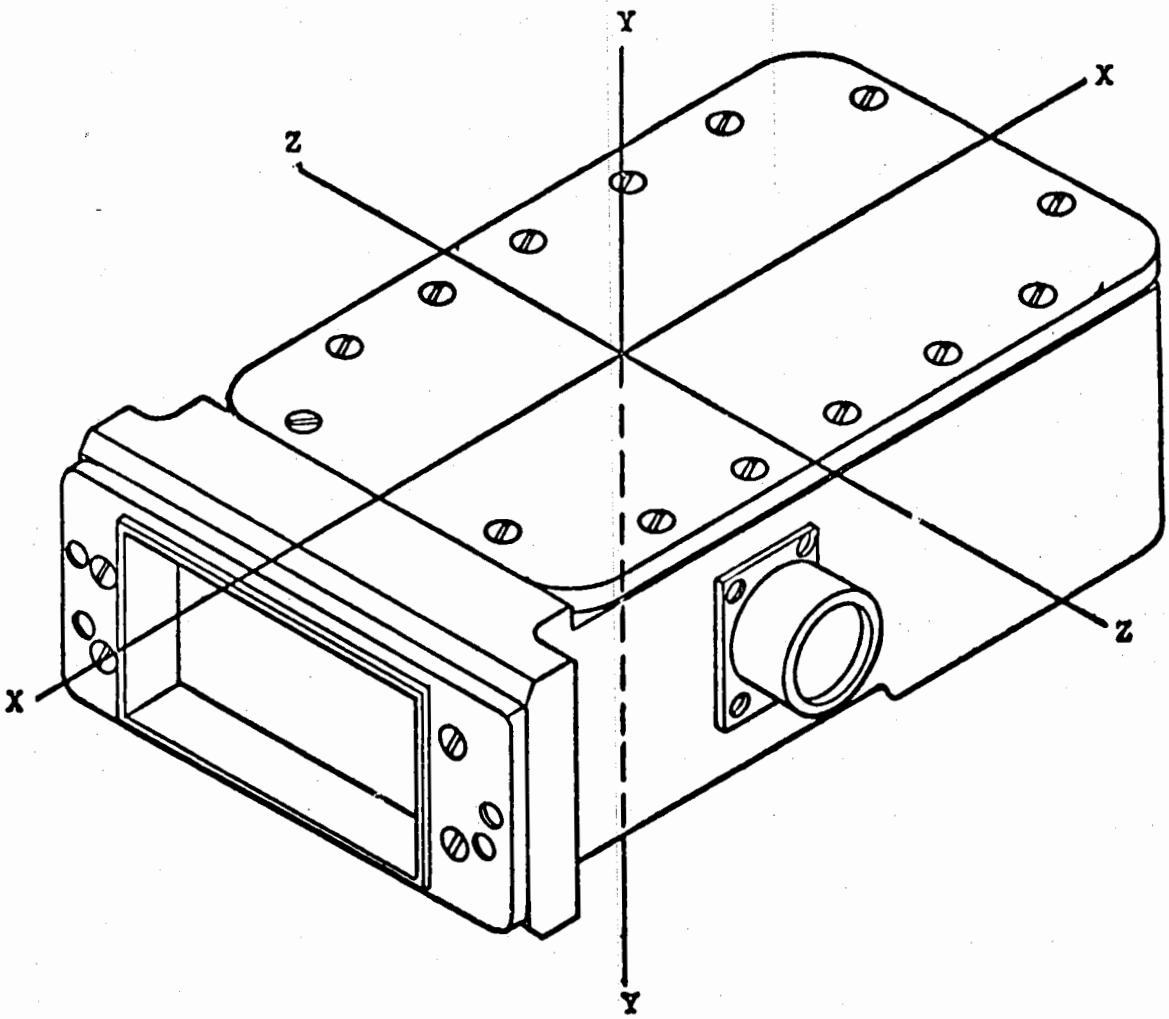
6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2261 Rev.I dated 14 September 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A057

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.



SIMULATOR ASSEMBLY,
WARHEAD SECTION:
M: 3E3

Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

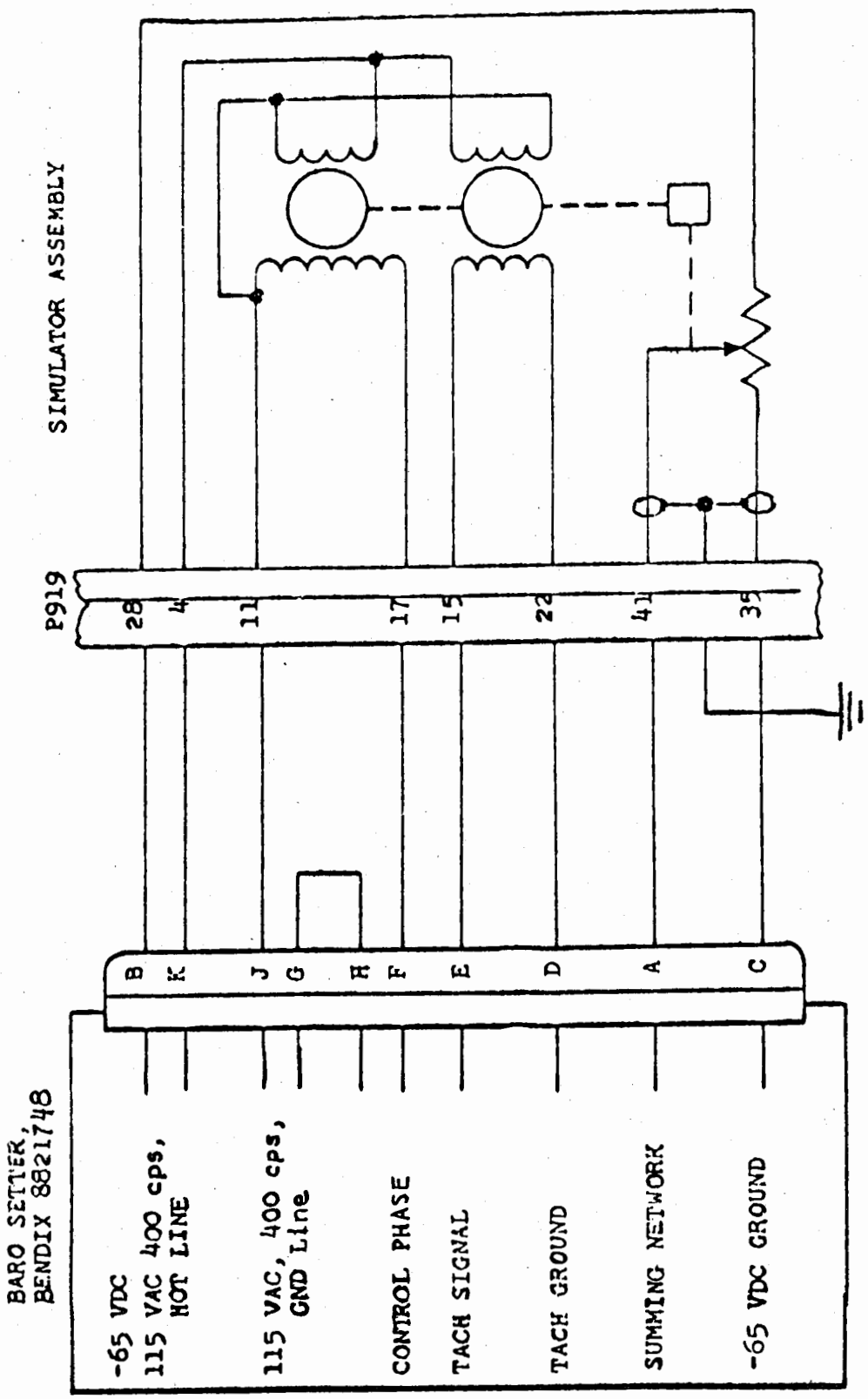


Figure 2. Simulator Test Circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.